

WAX — PROJECT FOUNDATION & RESEARCH STARTING DOCUMENT

A conceptual reference for beginning the project research independently. This is intentionally not an implementation blueprint. It records the ideas, philosophy, constraints, questions, analogies, and research direction developed so far.

Status: Foundation / pre-research / pre-architecture

Purpose: This document is a starting memory for the project. It should remain useful whether the next research session is conducted with ChatGPT, another AI, a research agent, or independently.

1. WHY THIS DOCUMENT EXISTS

An idea becomes dangerous to lose when it is bigger than the current codebase. WAX has reached that point. The purpose of this document is therefore not to tell an engineer exactly what to code. It is to preserve the conceptual direction before technology, schemas, frameworks, cloud providers, or current limitations begin defining the project by accident.

The project should be researched from first principles. Research should establish what is technically possible, what existing systems already demonstrate, what abstractions are actually necessary, what security boundaries are required, and what tradeoffs exist. Only after that should architecture be stabilized and implementation begin.

This document is deliberately closer to a notebook, constitution seed, research charter, and conceptual compass than a conventional product specification.

2. THE PROJECT'S ORIGIN AND EVOLUTION

WAX began as an educational AI idea: a WhatsApp AI tutor for Nigerian students, initially focused on JSS1–SS3 and examinations such as WAEC, NECO, JAMB, and BECE. The original goal was meaningful: help students understand rather than merely receive answers.

Over time, repeated implementation attempts exposed a deeper problem. The system could be made technically functional while still feeling scripted. Even after prompt improvements, the AI could fall into a funnel: ask the subject, ask the topic, subdivide the topic, then teach. That behavior was not merely a wording problem. The surrounding architecture had already defined the world as student + subject + topic + lesson + tutoring flow.

The resulting realization is that the project may be asking the wrong architectural question. Instead of designing a better tutor, the project is investigating whether it can create a general-purpose environment in which an intelligence can pursue human objectives, with education as its first and most important mission but not its permanent boundary.

3. CORE PHILOSOPHY — NEWBORN AI

The phrase “Newborn AI” originally meant that the intelligence should not be born into a rigid script. The deeper interpretation is now: do not predefine the entire world in which the intelligence must operate.

The AI should not be forced into predetermined subjects, topics, exams, lessons, tutoring modes, onboarding funnels, greetings, fixed workflows, or artificial conversational stages. It should encounter the human's objective and reason about what is needed.

Newborn does not mean unintelligent or empty. The intelligence may have powerful general reasoning, knowledge, memory mechanisms, and access to tools. What should remain open is the mapping between a human objective and the means used to pursue it.

A useful principle is: the system should provide an environment and mechanisms; the intelligence should determine how those mechanisms are composed to pursue legitimate objectives.

4. INFRASTRUCTURE, NOT INTELLIGENCE

The original WAX principle “Infrastructure, not intelligence” remains central. The infrastructure should not try to become the tutor, teacher, planner, or personality. It should provide the conditions under which intelligence can operate reliably.

The core infrastructure may necessarily contain deterministic mechanisms for security, identity, authorization, persistence, networking, isolation, resource management, delivery, observability, audit, recovery, and similar foundational concerns. These are not the same as hardcoded business logic.

The critical boundary is therefore not “no code may contain concepts.” The boundary is: universal infrastructure should not secretly encode assumptions about the domain or about what humans will ask the AI to do.

5. THE SCHOOL AND TEACHER ANALOGY

Imagine building a school. The builder creates the physical environment but does not fill every classroom with every possible object or write a script for every lesson.

A teacher arrives. A student arrives with an objective. The teacher determines what is required to accomplish that objective. If a marker is needed and available, the teacher uses it. If it is not available but a supplier exists, the teacher can seek it from the supplier. If the supplier does not have it, the teacher may look for another permitted route, such as obtaining it externally. The important part is that nobody had to hardcode a “marker workflow.”

The analogy captures the desired relationship: environment provides possibilities and constraints; intelligence interprets objectives and decides how to use, combine, discover, obtain, or request what is necessary.

The school is not the teacher. A supplier is not the teacher. The internet is not the teacher. A marker is not the teacher. They are resources or mechanisms in the teacher's environment.

6. THE AIRPLANE AND PILOT ANALOGY

Another central analogy is an airplane. An aircraft should not be architecturally defined as “the machine that flies one route.” The route is an objective. The aircraft, instruments, communication systems, navigation, fuel, and surrounding infrastructure make many legitimate journeys possible.

The pilot receives an objective and determines how to achieve it using what is available. If a direct route is unavailable, the pilot evaluates alternatives and constraints.

For WAX, the AI is analogous to the pilot. The runtime/environment is analogous to the aircraft and its operating systems. The underlying cloud, machines, networks, storage, and compute are analogous to the physical infrastructure. Interfaces such as WhatsApp and web are ways the human interacts with the system, not the definition of the intelligence itself.

The airplane analogy reinforces a crucial rule: design for a world of possible objectives, not for one predetermined route.

7. OPEN-WORLD WAX

“Open world” is one of the strongest descriptions of the emerging vision. It means that WAX should not be built around a closed catalogue of expected user activities.

A user might ask for help understanding biology, then ask for a document, then work on software, then ask for research, then request an external service, then move to a richer interface. The system should not need to become a different product for each objective.

Open world does not mean literally unlimited or magical. It means the system is designed to pursue arbitrary legitimate objectives within the capabilities, permissions, resources, and external connections that can safely be made available.

The important question is not “Did we program this exact use case?” but “Can the intelligence determine a route to the objective from the environment it has?”

8. CAPABILITY GAPS — UNAVAILABLE IS NOT THE SAME AS IMPOSSIBLE

A central concept is the difference between “the system cannot do this” and “the system cannot currently do this with what it has.”

If an objective requires something unavailable, the intelligence should be able to reason about the gap. Possible outcomes include using an existing route, composing available mechanisms, discovering an external service, connecting an authorized service, provisioning an appropriate environment, requesting permission, asking the human to perform a required step, delegating to a human or service, or concluding that the objective cannot safely or legitimately be completed.

The system should not treat a missing capability as an automatic failure without investigating whether the capability can be obtained or whether another route exists.

However, “make unavailable available” must never mean unrestricted autonomy. Acquisition and expansion remain subject to authorization, security, resource, legal, and safety boundaries.

9. EXTERNAL SERVICES AND THE WORLD OUTSIDE WAX

WAX should be conceptually capable of interacting with an ecosystem beyond itself. Examples discussed include Spotify, Gmail, GitHub, Google services, the web, developer services, marketplaces, and other external systems.

If a human asks for something that requires an external service, WAX should reason about which route is appropriate. The route might be an API, browser interaction, application integration, device integration, or another authorized mechanism. The AI should not be permanently tied to one integration method.

The larger principle is that the environment can extend into the external world through controlled connections. Specialized services should not have to become permanent assumptions inside the universal intelligence core.

10. HUMAN SERVICE DISCOVERY AND DELEGATION

The open-world idea can extend beyond software APIs. Suppose a user needs a web developer. A sufficiently connected WAX environment could search an opted-in directory, marketplace, or service network, identify suitable people or organizations, and—where authorized—help contact or engage them.

This is an example of a broader principle: when the objective cannot or should not be completed solely by the AI, the intelligence can discover an external human or service that can contribute.

This should be researched carefully because identity, consent, privacy, fraud prevention, contracts, payment, reputation, and accountability become important. The concept is exploratory, not a declaration that WAX must immediately build a human marketplace.

11. INTERFACES ARE WINDOWS, NOT THE INTELLIGENCE

The first important interfaces are expected to be WhatsApp and web. WhatsApp is valuable for natural conversation and accessibility. Web is valuable when the work requires richer interaction, larger information surfaces, files, editing, coding, browser interaction, or other capabilities that are awkward in messaging.

The conceptual goal is not to build separate intelligences for WhatsApp and web. They should be interfaces into the same underlying intelligence and continuity.

Telegram, mobile applications, desktop interfaces, voice, and future interfaces can be added later without changing the fundamental identity of the intelligence.

A user should ideally experience this as one continuing relationship rather than as separate products.

12. DYNAMIC INTERFACE TRANSITION

The AI should understand the delivery and interaction surfaces available to it at the current moment.

If a user is working on something that fits naturally in WhatsApp, remain there. If the objective becomes too large or interactive for WhatsApp and a richer web environment is available, the intelligence can recognize the mismatch and propose a transition.

If Telegram is available and useful, WAX can ask whether the user wants to continue there. A secure transition mechanism could take the user into the corresponding conversation while preserving continuity.

The deeper principle is not “build a Telegram handoff.” It is: the intelligence should be aware of the communication surfaces available to it and choose an appropriate surface for the work, subject to authorization and user control.

13. TEMPORARY RICHER ENVIRONMENTS

A major future possibility is temporary execution environments. A user may begin in WhatsApp, then WAX may determine that the objective requires a richer environment where files, terminals, code, browser interaction, large outputs, or other resources are practical.

WAX could potentially provision or expose an appropriate environment for the user, associate it with the ongoing work, and allow the intelligence to continue operating there. The environment might be temporary rather than a permanent account-level concept.

This suggests a research area around sandboxes, containers, ephemeral environments, browser execution, object storage, compute allocation, resource isolation, and secure user access.

The exact terminology and internal representation should not be fixed now. “Workspace” is an example of a possible experience, not a declaration of WAX’s universal ontology.

14. MEMORY AND CONTINUITY

Memory should serve continuity rather than force a predetermined student profile. The intelligence should be able to preserve useful information about a human and ongoing work across interruptions, sessions, and interfaces.

A person might spend an hour working with WAX, leave, and return later. The system should be capable of reconstructing the relevant state and continuing rather than behaving as if the previous hour never existed.

The project should research long-term memory, short-term context, episodic information, semantic information, summarization, compression, retrieval, conflict handling, confidence, supersession, and forgetting without assuming that the user is fundamentally a “student.”

One promising research question raised in discussion is whether a smaller supporting model or process could periodically determine what should be retained after a run, rather than relying entirely on the primary model to remember to save everything. This is a research hypothesis, not a fixed design.

15. LONG-RUNNING OBJECTIVES AND RESUMABILITY

WAX should be researched as an intelligence capable of continuing work beyond one response. A user objective may require many steps, external calls, waiting, retries, inspection, correction, or continuation after interruption.

The system therefore needs conceptual answers for durable state, checkpoints, resumability, idempotency, recovery, execution history, partial completion, and user-visible progress.

A critical principle is that an interrupted process should resume intelligently rather than blindly regenerate an answer or repeat completed actions.

16. SECURITY — A POWERFUL AI NEEDS A PHYSICAL LAW

The more open the environment becomes, the more important security becomes. The AI should have agency but not sovereignty.

The environment must enforce boundaries around identity, authentication, authorization, isolation, secrets, external accounts, sensitive data, resource limits, execution privileges, network access, auditability, revocation, and user consent.

The intelligence can request an action or capability. The environment decides whether that request is permitted. The AI should not be able to redefine the rules that constrain it.

Prompt injection, malicious external content, compromised integrations, credential theft, data exfiltration, tool abuse, privilege escalation, supply-chain risk, and unsafe autonomous actions must all become dedicated research areas.

17. THE BIG RULE ABOUT HARD-CODING

The project is not trying to eliminate every deterministic component. That would be impossible and undesirable.

The rule is to prevent business-domain assumptions from becoming invisible architecture. A database driver can be deterministic. Authentication can be deterministic. A WhatsApp adapter can be deterministic. A security boundary can be deterministic.

What should not be hardcoded into the universal intelligence environment is a worldview such as: every human is a student; every conversation has a subject; every objective is a lesson; every interaction follows onboarding; every task has one predetermined workflow; every response has a fixed length; or every useful capability must be known in advance.

Avoid replacing one rigid ontology with a new supposedly universal list of objects. Concepts such as workspace, task, artifact, event, capability, project, or goal may eventually be useful implementation constructs, but research must determine their necessity rather than declaring them the universe in advance.

18. EDUCATION REMAINS THE HEART

The project has not abandoned education. The original and most meaningful mission remains helping students understand. WAX should be exceptionally good at learning, explanation, reasoning, practice, research, and academic support.

The difference is that education is treated as the first purpose rather than the boundary of the environment.

This allows WAX to meet a student where they actually are. A student may ask a school question, then need a document, research, coding help, an explanation, a calculation, a file transformation, or an external service. The

system should not force them back into a tutoring menu.

19. THE HUMAN EXPERIENCE

The desired experience is conversational rather than kiosk-like. A human should not have to learn WAX's internal categories before getting useful help.

The AI should discover what the person needs. It should avoid unnecessary menus, forced modes, artificial onboarding, repetitive clarification, and unsolicited workflows.

A clarification question should be asked because it materially helps, not because a flowchart requires it.

The system should be capable of remembering meaningful preferences and context naturally, while avoiding invasive or unnecessary retention.

20. “THE AI DOES EVERYTHING” — THE PRECISE INTERPRETATION

The phrase means the intelligence should be responsible for deciding how to accomplish an objective, rather than infrastructure dictating a fixed workflow.

It does not mean the model directly controls every low-level machine component. The runtime must mediate access. The AI reasons over what is possible; the environment safely exposes what can actually be done.

The intelligence should be able to choose among available routes rather than being programmed with one route for each use case.

This is the distinction between intelligence using an environment and infrastructure pretending to be intelligence.

21. MODEL / RUNTIME SEPARATION

The model should be replaceable. A particular provider, model family, or current AI vendor should not become WAX's identity.

Different models may eventually serve different roles, and model selection can change as technology changes. The orchestration layer should therefore treat intelligence as a resource that can be routed and composed rather than as a permanent vendor dependency.

The research should investigate multi-model systems, model routing, tool-use reasoning, planning, verification, specialized models, fallback behavior, cost/latency tradeoffs, and model failure modes.

22. SCALE WITHOUT DESIGNING AROUND TODAY'S RAILWAY

Railway is a useful current deployment and testing environment. It should not become the definition of WAX's architecture.

The long-term environment may require multiple compute classes, persistent storage, object storage, GPU resources, browser execution, isolated containers, networked services, queues, orchestration, regional deployment, and other infrastructure.

Current budget limitations should not be allowed to force permanent architectural assumptions. Economic controls, credits, subscriptions, or resource tiers can be researched later. The immediate conceptual task is to understand the architecture that would make the vision possible.

23. WHAT WAX IS NOT

WAX is not merely a WhatsApp bot.

WAX is not merely a tutoring script.

WAX is not a collection of hardcoded educational workflows.

WAX is not a fixed list of supported use cases.

WAX is not a collection of interfaces pretending to be one AI.

WAX is not unrestricted autonomous software with root access to everything.

WAX is not defined by PostgreSQL, Redis, Railway, WhatsApp, one model provider, or any current technology.

WAX is not required to predict every future human objective in advance.

24. WHAT WAX IS TRYING TO BECOME

A general-purpose intelligent environment in which an AI can understand human objectives, reason about what is required, use what is available, discover or obtain additional permitted means when necessary, interact with external systems and people, preserve continuity, move between interfaces, and recover from interruption—while operating inside strong security, authorization, privacy, and resource boundaries.

Its first mission is education. Its architectural ambition is broader.

25. RESEARCH-FIRST RULE

No major implementation should be allowed to define the project before the concept is understood.

Research should come before architecture stabilization. Architecture should come before serious implementation. Implementation should be evaluated against the philosophy rather than used to define the philosophy.

Research outputs should be saved as independent documents so that the project remains understandable even if the current AI assistant, coding agent, repository, model provider, or development environment changes.

26. RESEARCH QUESTIONS TO CARRY FORWARD

What existing systems most closely resemble a general-purpose AI runtime or agent operating environment?

What is the minimum environment an intelligence needs to act meaningfully in the digital world?

How can an AI discover capabilities without being born with a closed catalogue?

How can new capabilities be safely connected, composed, acquired, provisioned, or revoked?

How can external services be integrated dynamically through APIs, browsers, applications, devices, or other mechanisms?

How can long-running objectives persist across failures and interfaces?

How should memory be represented, consolidated, retrieved, corrected, superseded, and forgotten?

How can temporary compute and execution environments be created safely?

How can a human's identity and continuity persist across WhatsApp, web, Telegram, mobile, desktop, and future interfaces?

How can the AI choose an appropriate interface for a particular objective?

How can the system distinguish capability absence from true impossibility?

How can security preserve AI agency without granting unrestricted sovereignty?

How can prompt injection and hostile external environments be contained?

What distributed-systems mechanisms are required for idempotency, retries, consistency, observability, and recovery?

What should be universal infrastructure, and what should remain outside the core?

Which abstractions are genuinely necessary, and which would merely be premature ontology?

How can education remain the first-class mission without turning the entire environment into a tutor-shaped machine?

What existing research, open-source projects, standards, protocols, and commercial systems have already solved pieces of this problem?

What would a truly model-independent runtime require?

What would it take to scale from a prototype to a global computing environment without rewriting the conceptual foundation?

27. PRINCIPLES TO PROTECT DURING FUTURE RESEARCH

Do not let current technology dictate the vision.

Do not replace student-specific hardcoding with a different rigid universal ontology.

Do not assume that every future capability must be known today.

Do not confuse an interface with the intelligence.

Do not confuse infrastructure with intelligence.

Do not confuse “autonomous” with “unrestricted.”

Do not turn education into a boundary when it is the first mission.

Do not optimize the entire architecture around current budget constraints.

Do not start coding merely because the idea feels exciting.

Do not let an implementation agent silently make foundational philosophical decisions.

Do not treat a successful prototype as proof that the underlying architecture is correct.

Do not accept scripted UX merely because it technically works.

Do not create permanent concepts merely because they are convenient database tables.

28. THE CENTRAL QUESTION

Everything can ultimately be reduced to one research question:

What would have to be true for an AI to operate as a general-purpose intelligence inside a secure, persistent, extensible digital environment, where it can pursue legitimate human objectives without requiring the builders to predict and hardcode those objectives beforehand?

That is the question the research should answer before WAX's final architecture is frozen.

29. A WORKING NORTH STAR

WAX should feel less like opening a specialized application and more like gaining access to an intelligent environment.

The human brings an objective. The intelligence understands it. The environment reveals what is currently possible. The intelligence determines what is needed. It uses available means, searches for missing means, asks for permission when necessary, obtains or composes additional means when legitimately possible, acts, observes the result, remembers what matters, recovers when things fail, and continues until the objective is complete or a genuine boundary is reached.

The human should not need to understand the machinery underneath this process.

That is the experience we are trying to investigate.

30. TOMORROW'S NEXT STEP

The next stage is not to begin coding. It is to turn this foundation into a disciplined research program.

The eventual research should be divided into focused sections, with each section producing a saved research document. Those documents should gather evidence, existing approaches, technical possibilities, failure modes, security considerations, open questions, and implications for WAX.

After the research has matured, the material can be compiled into a much more authoritative WAX constitution / architecture doctrine. Only then should implementation planning become detailed enough for an engineering agent to act on it.

The project should remain understandable even without the original conversation. Any future AI should be able to read the saved research and understand why WAX exists, what it is trying to become, what assumptions it must avoid, and what questions remain unanswered.